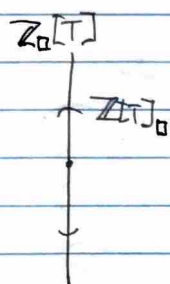


1. The analytic ring $\mathbb{Z}[T]_0$.

1.1. We have seen the analytic ring $\mathbb{Z}[T]$, where $M \in \mathcal{D}(\mathbb{Z}[T])$ is complete if M is complete as a $\mathbb{Z}$ -module (i.e. solid). One should think of $\mathbb{Z}$ as the base for non-arch. geo., and $\mathbb{Z}[T]$ the affine line.

We now introduce an analytic ring $\mathbb{Z}[T]_0$, which models the adic unit disc.



The first key point is that the adic unit disc will be modeled still by $\mathbb{Z}[T]_0$ but with a different analytic structure, and one should expect $\mathbb{Z}[T]_0 \otimes_{\mathbb{Z}} \mathbb{Q}_p \approx \mathbb{Q}_p\langle T \rangle$, the classical model for the adic unit disc $\mathbb{D}^{\text{ad}}$ over $\mathbb{Q}_p$.

The second key point is that the feature of $\mathbb{Q}_p\langle T \rangle$ to be taken as the hint for defining $\mathbb{Z}[T]_0$ is the following:
 $\forall$ nullseq. $(a_n) \in \mathbb{Q}_p$, $\sum_{n \geq 0} a_n T^n$ exists in $\mathbb{Q}_p\langle T \rangle$.

Def $\mathbb{Z}[T]_0 := (\mathbb{Z}[T], \mathcal{D}(\mathbb{Z}[T]_0))$, where $M \in \mathcal{D}(\mathbb{Z}[T])$ is complete if M is solid and every nullseq. in M is T -summable. More precisely: let $P' = P \otimes_{\mathbb{Z}} \mathbb{Z}[T] \approx \mathbb{Z}[T][T]$, then M is complete (T-solid) if

$$\text{Hom}_{\mathbb{Z}[T]}(P', M) \xrightarrow{1-T} \text{Hom}_{\mathbb{Z}[T]}(P', M) \text{ is an iso.}$$

Let's play with this definition:

$$\text{have } 0 \rightarrow \mathbb{Z}[T][T] \xrightarrow{1-T} \mathbb{Z}[T][T] \rightarrow \mathbb{Z}((T^{-1})) \rightarrow 0$$

as $\mathbb{Z}[T]$ -modules.

Discussion. This kind of statements can be checked in the ^{classical} world of top. ab. gps, using the following:

Lemma Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be a strictly exact seq. of locally compact abelian groups. Then the induced seq. of condensed abelian groups is exact.

So $M \in \mathcal{D}(\mathbb{Z}[T])$ is T-solid if M is solid and $\text{Hom}_{\mathbb{Z}[T]}(\mathbb{Z}((T^{-1})), M) = 0$.

One checks that $\mathbb{Z}((T^{-1}))$ is an idempotent $\mathbb{Z}[T]$ -algebra so we get an idempotent decomposition:

$$D(\mathbb{Z}((T^{-1}))) \xleftrightarrow{i^*} D(\mathbb{Z}_0[T]) \xleftrightarrow{j^*} D(\mathbb{Z}[T]_0)$$

$\xleftarrow{j!} \quad \xleftarrow{i_*}$

One can write down formulas for these functors by the general statements for idempotent completions.

$$\begin{array}{c} \mathbb{Z}_0[T] \\ \downarrow \\ \mathbb{Z}((T^{-1})) \\ \downarrow \\ \mathbb{Z}[T]_0 \end{array}$$

Geometrically, M on $\mathbb{Z}_0[T]$ is T -solid if and only if it comes from $(*, \text{ext.}) \mathbb{Z}[T]_0$.

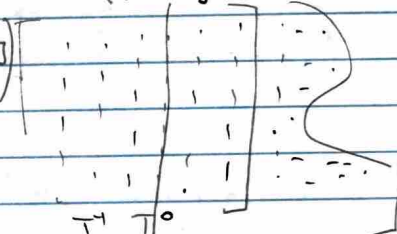
Thm. $\mathbb{Z}[T]_0$ is an analytic ring.

For $S = \varprojlim_n S_n$ in $\text{ProFin}^{\text{op}}$, (left) $\mathbb{Z}[T]_0[S] \simeq \varprojlim_n \mathbb{Z}[T][S_n]$.

We will not prove this, except mention that the key computation is:

$$0 \rightarrow \varprojlim_n \mathbb{Z}[T] \rightarrow \varprojlim_n (\mathbb{Z}[T]) \rightarrow \varprojlim_n (\mathbb{Z}((T^{-1}))) / (\varprojlim_n \mathbb{Z}_0 \otimes \mathbb{Z}((T^{-1}))) \rightarrow 0$$

In particular, $\left(\varprojlim_n \mathbb{Z}[T] \right)_{\mathbb{Z}_0[T]} \otimes_{\mathbb{Z}_0[T]} \mathbb{Z}[T]_0 \simeq \varprojlim_n \mathbb{Z}[T]$



1.2 We wanted the base change of $\mathbb{Z}[T]_0$ over $\mathbb{Z}_0$ to over $\mathbb{Q}$ be the aff. univ. disc. over $\mathbb{Q}$. Let's verify this.

First we need to make this precise, namely, to make an adic space an analytic stack.

Let $(A^\flat, A^+)$ be a complete Huber pair. Its associated analytic ring is defined as follows.

The underlying condensed ring is $A^\flat$ (clearly, $A^\flat$ is solid).

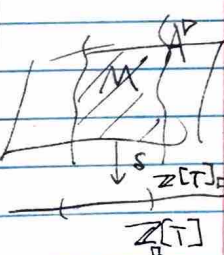
$M \in D(A^\flat)$ is complete if M is solid and $\forall s \in A^+$, inducing $\mathbb{Z}[T] \rightarrow A^\flat, T \mapsto s$, M is T -solid.

Denote $(A^\flat, A^+)_\square$ this analytic ring.

E.g. $A^\flat \in D(A^\flat, A^+)_\square$: $\forall s \in A^+$, every nullseq. $(a_n) \in A^\flat$, have $\sum_{n \geq 0} a_n s^n \in A^\flat$ because s is bounded so $(a_n s^n)$ is null.

Thus (Andreyekher) $(A^\flat, A^+)_\square$ is an analytic ring. The functor $\{\text{complete Huber pair}\} \rightarrow \{\text{analytic rings}\}$ is fully faithful.

Prop (loc. cit.) $(A^\flat, A^+)_\square \otimes_{\mathbb{Z}_0} \mathbb{Z}[T]_0 \simeq (A^\flat(T), A^+(T))_\square$



Ex $(\mathbb{Q}_p, \mathbb{Z}_p)_{\mathbb{Z}_0} \otimes_{\mathbb{Z}_0} \mathbb{Z}[T]_{\mathbb{Z}_0} \simeq (\mathbb{Q}_p\langle T \rangle, \mathbb{Z}_p\langle T \rangle)_{\mathbb{Z}_0}$.

Let's at least verify $\mathbb{Q}_p \otimes_{\mathbb{Z}_0} \mathbb{Z}[T]_{\mathbb{Z}_0} \simeq \mathbb{Q}_p\langle T \rangle$ as $\mathbb{Z}[T]_{\mathbb{Z}_0}$ mod

$\mathbb{Z}_p \simeq \varprojlim_n \mathbb{Z}/p^n$. ~~so $\mathbb{Z}_p \otimes$~~

Fact $(-) \otimes_{\mathbb{Z}_0} \mathbb{Z}[T]_{\mathbb{Z}_0} : D(\mathbb{Z}_0) \rightarrow D(\mathbb{Z}[T]_{\mathbb{Z}_0})$ commutes with limits. [To be discussed]

So $\mathbb{Z}_p \otimes_{\mathbb{Z}_0} \mathbb{Z}[T]_{\mathbb{Z}_0} \simeq \varprojlim_n (\mathbb{Z}/p^n[T]) \simeq \mathbb{Z}_p\langle T \rangle$ in §2

$\mathbb{Q}_p \simeq \oplus \mathbb{Z}_p$, so $\mathbb{Q}_p \otimes_{\mathbb{Z}_0} \mathbb{Z}[T]_{\mathbb{Z}_0} \simeq \oplus (\mathbb{Z}_p\langle T \rangle) \simeq \mathbb{Q}_p\langle T \rangle$.

2. Four functors for $f: \text{AnSpec}(\mathbb{Z}[T]_{\mathbb{Z}_0}) \rightarrow \text{AnSpec}(\mathbb{Z}_0)$

f_* is just forgetting. We want to describe $f^*, f^!, f_!$ explicitly, in particular, we want to show f^* commutes with limits. We will give an alt. We focus on f^* . The descriptions for $f^!$ & $f_!$ will follow.

Note: this is false in the classical schemes context!

Have $\text{AnSpec}(\mathbb{Z}[T]_{\mathbb{Z}_0}) \hookrightarrow \text{AnSpec}(\mathbb{Z}_0[T]) \xrightarrow{p} \text{AnSpec}(\mathbb{Z}_0)$, with j an open immersion, p proper. The idempotent decomposition for j gives $j_! \mathcal{V} \simeq \text{fib}(\mathbb{Z}[T] \rightarrow \mathbb{Z}((T^{-1}))) \otimes_{\mathbb{Z}_0[T]} \mathcal{V}$.

Explicitly, $\text{fib}(\mathbb{Z}[T] \rightarrow \mathbb{Z}((T^{-1}))) \simeq \mathbb{Z}[T]^{\vee}[-1]$, where $\mathbb{Z}[T]^{\vee} := \text{Hom}_{\mathbb{Z}[T]}(\mathbb{Z}[T], \mathbb{Z})$, a $\mathbb{Z}_0[T]$ -mod.

So $f^* \mathcal{M} \simeq j^* p^* \mathcal{M} \simeq \text{Hom}_{\mathbb{Z}_0[T]}(\mathbb{Z}[T]^{\vee}, \mathcal{M} \otimes_{\mathbb{Z}_0} \mathbb{Z}_0[T])[-1]$.

This is good, but does not show f^* comm. with limits. Remarkably:

Claim $f^* \mathcal{M} \simeq \text{Hom}_{\mathbb{Z}_0}(\mathbb{Z}[T]^{\vee}, \mathcal{M})$.

The map is induced by $\mathbb{Z}[T]^{\vee} \rightarrow \mathbb{Z}$. As $\mathbb{Z}[T]^{\vee} \simeq \prod \mathbb{Z}$ as a $\mathbb{Z}_0$ -module, hence compact, it suffices to prove this for $\mathcal{M} = \prod \mathbb{Z}$, the compact generator for $D(\mathbb{Z}_0)$, i.e., we want:

$(\prod \mathbb{Z}) \otimes_{\mathbb{Z}_0} \mathbb{Z}[T]_{\mathbb{Z}_0} \simeq \prod \text{Hom}_{\mathbb{Z}_0}(\prod \mathbb{Z}, \mathbb{Z}) \simeq \prod (\mathbb{Z}[T])$.

This we know from §1.

$\text{Hom}_{\mathbb{Z}_0}(\prod \mathbb{Z}, \mathbb{Z}) \simeq \text{Hom}_{\mathbb{Z}}(\prod \mathbb{Z}, \mathbb{Z}) \simeq G(\prod \mathbb{Z}, \mathbb{Z})$
 $\text{Hom}_{\mathbb{Z}}(\prod \mathbb{Z}, \mathbb{Z}) \simeq G(\text{Aut}(\prod \mathbb{Z}), \mathbb{Z}) \simeq \oplus \mathbb{Z}$.

Consequently, $f^* \simeq \text{Hom}(\mathbb{Z}[T]^v, -)$ commutes with limits, and has the $\mathbb{Z}_0$ left adjoint $f_! \simeq (-) \otimes_{\mathbb{Z}_0[T]} \mathbb{Z}[T]^v$.

So $f_! \simeq p_* j_! \simeq f_![-1] \xrightarrow[\text{adj.}]{\text{right}} f^! \simeq f^*[1]$.

We have shown:

~~Prop~~ f is square and $f^! \mathbb{1} \simeq \mathbb{1}[1]$.

3. Recollection on étale and prim maps

Let (C, E) be a geometric setup, $f \in E$, D a $\mathbb{G}$ -fun. formalism. In one sentence, f being étale (resp. prim) ~~is a proper~~ means f ~~is~~ smooth (resp. proper) map when seen through D behaves like a the lens of the cohomology theory D .

To motivate its definition, consider the setup of ℓ -adic sheaves on alg. var.

Fact $A \in D(X)$ is locally constant constructible $\Leftrightarrow A$ is dualizable, i.e. $\exists A^\vee \in D(X)$ and ev: $A^\vee \otimes A \rightarrow \mathbb{1}$, coev: $A \otimes A^\vee \rightarrow \mathbb{1}$ s.t.
 $A \xrightarrow{\text{coev}} A \otimes A^\vee \otimes A \xrightarrow{\text{ev}} A$ is id_A & similarly for $A^\vee$.

The dualizability condition can be rephrased as follows:

$$\begin{array}{ccc} D(X) & \xrightarrow{\exists} & \\ A \otimes H(-) \uparrow & & \uparrow A' \otimes (-) \\ & D(X) & \end{array}$$

This suggests a relative version of dualizability (hence of being loc. const. constr. relative to a base):

$$f: X \rightarrow S, A \in D(X), f_!(A \otimes H(-)) \xrightarrow{\exists} f_!(A' \otimes f^*(-))$$

$D(S)$

Def Let (C, E) be a geo. setup, D a $\mathbb{G}$ -fun. formalism. $S \in C$. ~~Consider the~~ Let $(C_E)_S$ be the ℓ -able maps over S , D_S be the restriction of D to $(C_E)_S$.

Consider the 2-category $\text{Corr}_{D_S}((C_E)_S)$ (usually denoted by K_{D_S}):
 objects the same as $(C_E)_S$.
 morph. from X to $Y := D(X \overset{*}{\otimes} Y)$ (denoted by $\text{Fun}_D(X, Y)$)

Let $f: X \rightarrow S$ in $(C_E)_S$, $A \in D(X)$. Then A is called f -étale if A is a left adjoint when viewed as an object in $\text{Fun}_D(X, S) \simeq D(X)$.

A map $f: X \rightarrow S$ is called étale if $\mathbb{1}_X$ is f -étale.

Rmk (1) Replace "left" by "right", we get the def. of f -prim and prim.

(2) Writing down explicitly what f -suaveness means, we see that being f -suave only involves $X \rightrightarrows S$, $X \otimes X$, and f^* , $f_!$, pr^* , $pr_!$ and Δ .

(3) A being f -suave implies $\exists A' \in D(X)$ s.t.
 $f_!(A \otimes (-)) \simeq A' \otimes f^*(-)$.

This implies $A' \otimes f^*(-) \simeq \text{Hom}_X(A, f^!(-))$.

In particular $A' \simeq \text{Hom}_X(A, f^! \mathbb{A}_S)$.

A' is called the suave dual of A .

Similarly for f -primness.

(4) Note we require A itself be a left adjoint, rather than the functor $f_!(A \otimes (-))$ is a left adjoint.

The former is stronger, ~~for exam~~ and more natural.

For example, one easily shows f -suaveness is preserved under base change.

However, in certain situations ~~these two are equal~~ the former is equal to the latter + DG-linearly.

Prop Let (C, E) be a geo. setup, D a presentable G -fm. Assume D satisfies the categorical Künneth formula. Then, $\forall S \in C$, $K_{D,S} \rightarrow \text{Pr}_{D(S)}^L$ is sym. mon. and 2-fully faithful.

In this situation, to check f is suave, it suffices to show $f_!$ has a DG-linear right adjoint.

4. Recall for $f: \text{AnSpec}(\mathbb{Z}[T]_{\square}) \rightarrow \text{AnSpec}(\mathbb{Z}_{\square})$, we showed $f^*_{[1]} \simeq f^!$. So $f_! \simeq f^*[1]$. By §3, this indeed shows f is suave. The goal of this section is to discuss the following:

Thm Let $f: X \rightarrow Y$ be a smooth map of n.g. an. va. / $\mathbb{A}_p$, then the induced map of analytic stacks is suave.

The statement is true for more general spaces.
We restrict to w.g. an. va. / $\mathbb{Q}_p$, i.e. adic spaces locally of top. fin. type over $\mathrm{Spa}(\mathbb{Q}_p)$, for simplicity.

Sketch of pf It suffices to treat the aff. case $\begin{matrix} B \\ \uparrow \\ A \end{matrix}$.

By [Huber96, Cor. 1.6.10, Prop. 1.7.1], there exists a presentation

$$\begin{array}{ccc} B & \leftarrow \alpha & \\ \uparrow & \nearrow & A\langle T_1, \dots, T_n \rangle \\ A & & \end{array} \quad \text{where } \alpha \text{ is } A\langle T_1, \dots, T_n \rangle \twoheadrightarrow A\langle T_1, \dots, T_n \rangle / (f_1, \dots, f_m) \cong B,$$

with $f_1, \dots, f_m \in A\langle T_1, \dots, T_n \rangle$ s.t.
 $\det(\frac{\partial f_i}{\partial T_j})_{1 \leq i, j \leq m}$ is a unit in B .

By [RC24, Thm 3.5.6], this ~~induces~~ is sent to the diagram $B_\square \cong A\langle T_1, \dots, T_n \rangle_\square / (f_1, \dots, f_m)$ ~~via~~ the functor

$$\begin{array}{ccc} & & \uparrow \\ & & A\langle T_1, \dots, T_n \rangle_\square \\ \uparrow & \nearrow & \\ A_\square & & \end{array}$$

$\{\text{complete Huber pair}\} \rightarrow \{\text{analytic rings}\}$

We want to show $A_\square \rightarrow A\langle T_1, \dots, T_n \rangle_\square$ is surjective and $A\langle T_1, \dots, T_n \rangle_\square \rightarrow A\langle T_1, \dots, T_n \rangle_\square / (f_1, \dots, f_m)$ is surjective.

As $A\langle T_1, \dots, T_n \rangle_\square \leftarrow \mathbb{Z}\langle T_1, \dots, T_n \rangle_\square$ is a pushout and pr is surjective, $A_\square \rightarrow A\langle T_1, \dots, T_n \rangle_\square$ is surjective.

$$\begin{array}{ccc} & \mathbb{Z}\langle T_1, \dots, T_n \rangle_\square & \\ \uparrow & \uparrow \mathrm{pr} & \\ A_\square & \leftarrow \mathbb{Z}_\square & \end{array}$$

As $A\langle T_1, \dots, T_n \rangle_\square / (f_1, \dots, f_m) \leftarrow \mathbb{Z}_\square$ is a pushout, the surjectiveness of the left vertical map is reduced to

$$\begin{array}{ccc} A\langle T_1, \dots, T_n \rangle_\square & \leftarrow & \mathbb{Z}\langle T_1, \dots, T_n \rangle \\ \uparrow f_i & & \uparrow i \\ A\langle T_1, \dots, T_n \rangle & \leftarrow & \mathbb{Z} \end{array}$$

the surjectiveness of the right vertical map, hence further to that of $\mathbb{Z}[T] \rightarrow \mathbb{Z}, T \mapsto 0$.

It is formal that one only needs to check $\mathbb{Z} \cong \mathbb{Z}[T]/T$ is surjective in $\mathcal{D}(\mathbb{Z}[T])$. This is easy:

$\mathrm{Hom}_{\mathbb{Z}[T]}(\mathbb{Z}, \mathbb{Z}[T]) \cong \mathbb{Z}[T]$, and one has the evident ev, coev.

□

5. Cartier duality.

The most classical Cartier duality is about commutative and co-commutative Hopf algebras. We will discuss a duality "one categorical level up", between $\mathcal{G}_A$ and $B\hat{\mathcal{G}}_A$. This will be useful later in the seminar, and also serves as an explicit example of computing with the 6-fun. formalism.

For lack of space/time, the plan is:

- (1) discuss $\mathcal{QCoh}(\mathcal{G}_A) \cong \mathcal{QCoh}(B\hat{\mathcal{G}}_A)$ in the algebraic context
- (2) state the analogues in the solid setting, for algebraic vector bundles and analytic vector bundles.
- (3) discuss in detail certain computations involving $\mathcal{G}_A, B\hat{\mathcal{G}}_A, B\mathcal{G}_A$, which are key inputs to proving (2).

5.1 $X := \mathrm{Spec}(\mathbb{Q})$. $\mathcal{G}_A := \mathrm{Spec}(\mathbb{Q}[X])$, $\hat{\mathcal{G}}_A := \mathrm{Spf}(\mathbb{Q}[[Y]])$.

Cartier duality states $\mathcal{D}_{\mathcal{G}_A}(\mathcal{G}_A) \cong \mathcal{D}_{\mathcal{G}_A}(B\hat{\mathcal{G}}_A)$.

Let's discuss this on the abelian level: $\mathcal{QCoh}(\mathcal{G}_A) \cong \mathcal{QCoh}(B\hat{\mathcal{G}}_A)$.

An object $M \in \mathcal{QCoh}(B\hat{\mathcal{G}}_A)$ is the data of a $\mathbb{Q}$ -mod. M equipped with a continuous $\mathcal{O}(\hat{\mathcal{G}}_A)$ -comodule structure $\alpha: M \rightarrow M \hat{\otimes} \mathbb{Q}[[Y]]$.

Write $\alpha: m \mapsto \sum \delta_i(m) Y^i$. Then each $\delta_i \in \mathrm{End}(M)$, and they satisfy the "comodule axioms", which amounts to $\delta_0 = \mathrm{id}$, $\delta_i \delta_j(m) = \delta_{i+j}(m) \binom{i+j}{i}$.

Since we are over $\mathbb{Q}$, $\delta_i \Leftrightarrow \delta_i = \delta_{i+1}/i!$. So the data of $M \in \mathcal{QCoh}(B\hat{\mathcal{G}}_A)$ is the same as (M, δ_1) , i.e. an object in $\mathcal{QCoh}(\mathcal{G}_A)$.

- Remarkably, this equivalence can be put in the form of a Fourier transform.

Consider $\mathcal{G}_A \times \hat{\mathcal{G}}_A \rightarrow \hat{\mathcal{G}}_A$, $\exp(XY) \leftarrow 1 + Y$. This is a group homomorphism when either of the two factors is fixed. So we get, in particular, a map $\mu: \mathcal{G}_A \times B\hat{\mathcal{G}}_A \rightarrow B\hat{\mathcal{G}}_A$. Let p (resp. q) be the projection $\mathcal{G}_A \times B\hat{\mathcal{G}}_A \rightarrow \mathcal{G}_A$ (resp. $B\hat{\mathcal{G}}_A$).

Def $F: \mathcal{QCoh}(\mathcal{G}_A) \rightarrow \mathcal{QCoh}(B\hat{\mathcal{G}}_A)$, $M \mapsto q_*(p^* M \otimes \mu^*(\mathcal{O}(1)))$.

- Let's verify that F coincides with the explicit equivalence above (i.e. $(M, X \in \mathrm{End}(M)) \mapsto (M \rightarrow M[[Y]], m \mapsto m + X(m)Y + X^2(m)Y^2 + \dots)$, ~~at least~~ (on the abelian level).

Exercise $\mu^*(\mathcal{O}(1)) \simeq (\mathbb{Q}[X], \text{multiplication}, f \mapsto f + XfY + \frac{X^2}{2!}fY^2 + \dots)$

[Note, an obj. $M \in \mathcal{Q}\text{Coh}(\mathbb{G}_a \times B\hat{\mathbb{G}}_a)$ is an $\mathcal{O}(\mathbb{G}_a)$ -module equipped with a continuous $\mathcal{O}(B\hat{\mathbb{G}}_a)$ -comod. structure which is $\mathcal{O}(\mathbb{G}_a)$ -linear.]

Given $(M, X \in \text{End}(M)) \in \mathcal{Q}\text{Coh}(\mathbb{G}_a)$, ~~$\mu^*(\mathcal{O}(1)) \simeq$~~
 $\mu^*M \simeq (M, X, m \mapsto m + 0 + 0 + \dots)$ (trivial $\mathcal{O}(\mathbb{G}_a)$ -comod. str.)
 $\mu^*M \otimes \mu^*(\mathcal{O}(1)) \simeq (M \otimes \mathbb{Q}[X], X, m \mapsto m \otimes f + (mf + m \otimes Xf)Y + \dots)$

$$\simeq (M, X, m (= m \otimes 1) \mapsto m + X(m)Y + \dots)$$

$p_*(\mu^*M \otimes \mu^*(\mathcal{O}(1)))$ forgets the second entry. ✓

[We are using: on $\mathbb{G}_a \times B\hat{\mathbb{G}}_a$, $(M, X, \delta) \otimes (N, X, \delta') \simeq (M \otimes N, X, \delta \otimes 1 + 1 \otimes \delta')$.]

The inverse to F is $p_*(f^*(-) \otimes \mu^*(\mathcal{O}(-1)))$ [1].

Indeed: $\mu^*(\mathcal{O}(-1)) \simeq (\mathbb{Q}[X], \text{mult.}, f \mapsto f - XfY + \frac{X^2}{2!}fY^2 - \dots)$

Given $(M, \delta \in \text{End}(M)) \in \mathcal{Q}\text{Coh}(B\hat{\mathbb{G}}_a)$,

$q^*M \simeq (M[X], \text{mult.}, m \otimes f \mapsto m \otimes f + (\delta(m) \otimes f)Y + \dots)$ $\simeq \sum \delta^i m_i X^i$

$q^*M \otimes \mu^*(\mathcal{O}(-1)) \simeq (M[X], \text{mult.}, m \otimes f \mapsto \sum m_i X^i \mapsto \sum m_i X^i + (-\sum m_i X^{i+1})$

$p_*(q^*M \otimes \mu^*(\mathcal{O}(-1))) \simeq \text{fib}(M[X] \rightarrow M[X])$, where

the map is $\sum m_i X^i \mapsto \sum (\delta^i m_i) X^i - m_i X^{i+1}$

Note these are polynomials in X , so

$$0 \rightarrow M[X] \rightarrow M[X] \rightarrow M' \rightarrow 0, \text{ where } M' \text{ is}$$

M as a $\mathbb{Q}$ -mod., $M[X] \rightarrow M'$ is $\sum m_i X^i \mapsto \sum \delta^i m_i$.

This implies the description of F^{-1} .

Rmk Denote $\mathbb{G}_a \times \mathbb{G}_a \xrightarrow{\hat{\alpha}} \mathbb{G}_a$. As in usual Fourier transform, one expects $\hat{\alpha}^* \xrightarrow{F} \hat{\alpha}$ F, F' interchanges ($\neq$!) push and pull functors. ! does not make sense in this context, but one can verify $p_* \simeq f^* F$, $\pi_* \simeq \hat{\alpha}^* F^{-1}[-1]$ etc.

5.2⁽¹⁾ Let $\mathbb{Q}$ be the discrete ring with induced analytic ring structure from $\mathbb{Z}$. In this section we work over $\mathbb{Q}$.

$\mathbb{G}_a := \text{AnSpec}(\mathbb{Q}[X])$, where $\mathbb{Q}[X]$ has the induced analytic ring str. from $\mathbb{Q}$. $\hat{\mathbb{G}}_a := \varinjlim_n \text{AnSpec}(\mathbb{Q}[Y]/Y^n)$, $\mathbb{Q}[Y]/Y^n$ with induced an. str.

[! $\hat{\mathbb{G}}_a$ is different from $\text{AnSpec}(\varinjlim_n \mathbb{Q}[Y]/Y^n) = \text{AnSpec}(\mathbb{Q}[[Y]])$.]

The latter is aff. while $\hat{\mathbb{G}}_a$ is NOT.

Have $\hat{\mathbb{G}}_a \rightarrow \text{AnSpec}(\mathbb{Q}[[Y]])$.

A similar exp map defines $\mu: \hat{G}_a \times B\hat{G}_a \rightarrow B\hat{G}_m$.

Thm $F := q_!(p^*(-) \otimes \mu^*(G(1)))$ and $G := p_!(q^*(-) \otimes \mu^*(G(1)))[-1]$ are inverse to each other, inducing $D(\hat{G}_a) \simeq D(B\hat{G}_a)$.

Furthermore, denote $\hat{G}_a \xrightarrow{p} * \xleftarrow{\pi} B\hat{G}_a$, we have

$$\begin{aligned} a^* &\simeq F!*, \quad \pi_* \simeq !^*G \\ \pi_! &\simeq !^*G[1], \quad \pi^! \simeq F!_*[-1] \end{aligned} \quad \begin{array}{l} p \text{ is prim, } \pi \text{ is square \& prim,} \\ \hat{G}_a \rightarrow * \text{ is square, } B\hat{G}_a \rightarrow * \text{ is square \& prim.} \end{array}$$

Prk The apparent discrepancy from §5.1 is from $p_! \simeq p_*[2]$.

(2). We now state a variant for analytic vector bundles. $* = \text{AnSpec}(\mathbb{Q}_p)$ ($\mathbb{Q}_p$ denotes $(\mathbb{Q}_p, \mathbb{Z}_p)_\square$, which is the same as the analytic ring str. induced from $\mathbb{Z}_\square$).

$$\hat{G}_a^{\text{an}} := \varinjlim_{n \rightarrow \infty} \text{AnSpec}(\mathbb{Q}_p[\frac{1}{p^n}T]_\square), \quad \hat{G}_a^+ := \varinjlim_{n \rightarrow \infty} \text{AnSpec}(\mathbb{Q}_p[\frac{1}{p^n}]_\square).$$

Note $\hat{G}_a^{\text{an}}$ is not aff., while $\hat{G}_a^+ = \text{AnSpec}(\varinjlim \mathbb{Q}_p[\frac{1}{p^n}]_\square)$.

Similarly, we have $\mu: \frac{\hat{G}_a^{\text{an}} \times \hat{G}_a^+}{\hat{G}_a^+ \times B\hat{G}_a^{\text{an}}} \rightarrow B\hat{G}_m$.

Thm Same statement as above, replacing $\hat{G}_a$ by $\hat{G}_a^+$, $B\hat{G}_a$ by $B\hat{G}_a^{\text{an}}$.

5.3 We discuss in detail some computations. Set-up: §5.2.1.

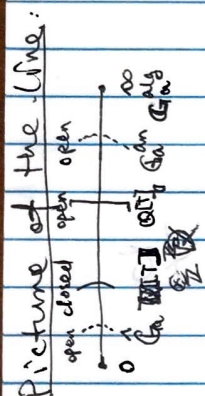
(1) $p: \hat{G}_a \rightarrow *$. As $\mathbb{Q}[X]$ has the induced analytic str., p is prim. $p_! \simeq p_* = \text{forget}$. p^* is just $- \otimes \mathbb{Q}[X]$, $p^! \simeq \text{Hom}(\mathbb{Q}[X], -)$.

[This is general: for $A \rightarrow B$ with B having the induced analytic str., $! \text{-pullback}$ is $\text{Hom}_A(B, -)$.]

In particular $p^!1_* \simeq \text{Hom}_{\mathbb{Q}}(\mathbb{Q}[X], \mathbb{Q}) \simeq \mathbb{Q}[X]^{\vee}$ ($\simeq \mathbb{Q}[X]$, X acts by $(a_0, a_1, \dots) \mapsto (a_1, a_2, \dots)$).

(2) $\hat{G}_a \xrightarrow{j} \hat{G}_a \xrightarrow{p} *$ j is an open immersion, with complement the idempotent algebra $\mathbb{Q}[X^+]$.

$\hat{p}$ is square: $\hat{G}_a \xrightarrow{\text{open}} \hat{G}_a^+ \xrightarrow{p^+} *$ has closed complement $\mathbb{Q}[X^+]$, $p^+ \rightarrow *$ is square, being covered by $\text{AnSpec}(\mathbb{Q}[X]_0)$ & $\text{AnSpec}(\mathbb{Q}[X^+]_0)$.



$$p' \simeq p^* \otimes p' \mathbb{1}.$$

Let's compute $p' \mathbb{1}$.

$$\begin{aligned} p' \mathbb{1} &\simeq j^* p' \mathbb{1} \simeq \underline{\text{Hom}}_{\mathbb{Q}[X]}(\text{fib}(\mathbb{Q}[X] \rightarrow \mathbb{Q}[X^{\pm 1}]), p' \mathbb{1}) \\ &\simeq \underline{\text{Hom}}_{\mathbb{Q}[X]}(X^{-1} \mathbb{Q}[X^{\pm 1}], p' \mathbb{1})[+1] \\ &\simeq \underline{\text{Hom}}_{\mathbb{Q}}(X^{-1} \mathbb{Q}[X^{\pm 1}], \mathbb{Q})[1] \\ &\simeq X \mathbb{Q}[X][1] \simeq \mathbb{1}[1]. \end{aligned}$$

So $p' \simeq p^* [1]$. $\hat{p}_* = \text{forget}$, $\hat{p}_! \simeq p_! j_! \simeq \text{fib}(\mathbb{Q}[X] \rightarrow \mathbb{Q}[X^{\pm 1}]) \otimes (-)$.

(3) $* \xrightarrow{f} B\hat{G}_a \xrightarrow{\pi} *$. Note f & π are $!$ -able. ([HM24, Thm 3.4.11])
([HM24, §4.5])

Facts. π -smoothness (resp. π -primness) can be checked after a universal π -cover of the target.

π -smoothness can be checked after a square universal π -cover of the source.

So f & π are π -smooth (using $\hat{p}$ square).

Compute $f' \mathbb{1}$: $\begin{array}{ccc} \hat{G}_a & \xrightarrow{\hat{p}^*} & * \\ \hat{p}' \downarrow \square \downarrow f & & \\ * & \xrightarrow{f} & B\hat{G}_a \end{array}$ $\hat{p}' f' \mathbb{1} \stackrel{\text{smooth base change}}{\simeq} \hat{p}'_! q' \mathbb{1} \simeq X \mathbb{Q}[X][1]$
So $q' \mathbb{1} \simeq \mathbb{1}[1]$ ($\hat{p}^*$ is conservative).
 $q' \simeq q^* [1]$.

Compute $\pi' \mathbb{1}$: $B\hat{G}_a \xrightarrow{\Delta} B\hat{G}_a \times B\hat{G}_a \xrightarrow{p_2} B\hat{G}_a$ $\pi' \mathbb{1} \simeq \Delta^* p_2^* \pi' \mathbb{1} \simeq \Delta^* p_1' \mathbb{1}$
 $\begin{array}{ccc} p_1 \downarrow \square \downarrow \pi & & \\ B\hat{G}_a & \xrightarrow{\pi} & * \end{array}$

Note $\hat{G}_a \rightarrow *$
 $\begin{array}{ccc} \hat{G}_a & \xrightarrow{\Delta} & B(\hat{G}_a \times \hat{G}_a) \\ \downarrow \square \downarrow & & \downarrow \square \downarrow \\ \hat{G}_a / \hat{G}_a \times \hat{G}_a & \xrightarrow{\Delta} & B\hat{G}_a \times B\hat{G}_a \end{array}$
 $\hat{G}_a / \hat{G}_a \times \hat{G}_a \simeq \hat{G}_a$
 $\Delta \simeq \Delta$

So Δ is square,
 $\Delta' \mathbb{1} \simeq \Delta^* \otimes \Delta' \mathbb{1}$
 $\Delta' \mathbb{1} \simeq \mathbb{1}[1]$. (computation similar as above)

So $\pi' \mathbb{1} \simeq \Delta' p_1' \mathbb{1} \otimes \mathbb{1}[-1] \simeq \mathbb{1}[+1]$. $\pi' \mathbb{1} \simeq \pi^* [1]$.